

Microservices Redesign of OPD-Vertex

Target scale: 50M users / 1M+ concurrent / ~50k consultations/min | p95 < 400 ms reads
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1. System Decomposition - 12 microservices

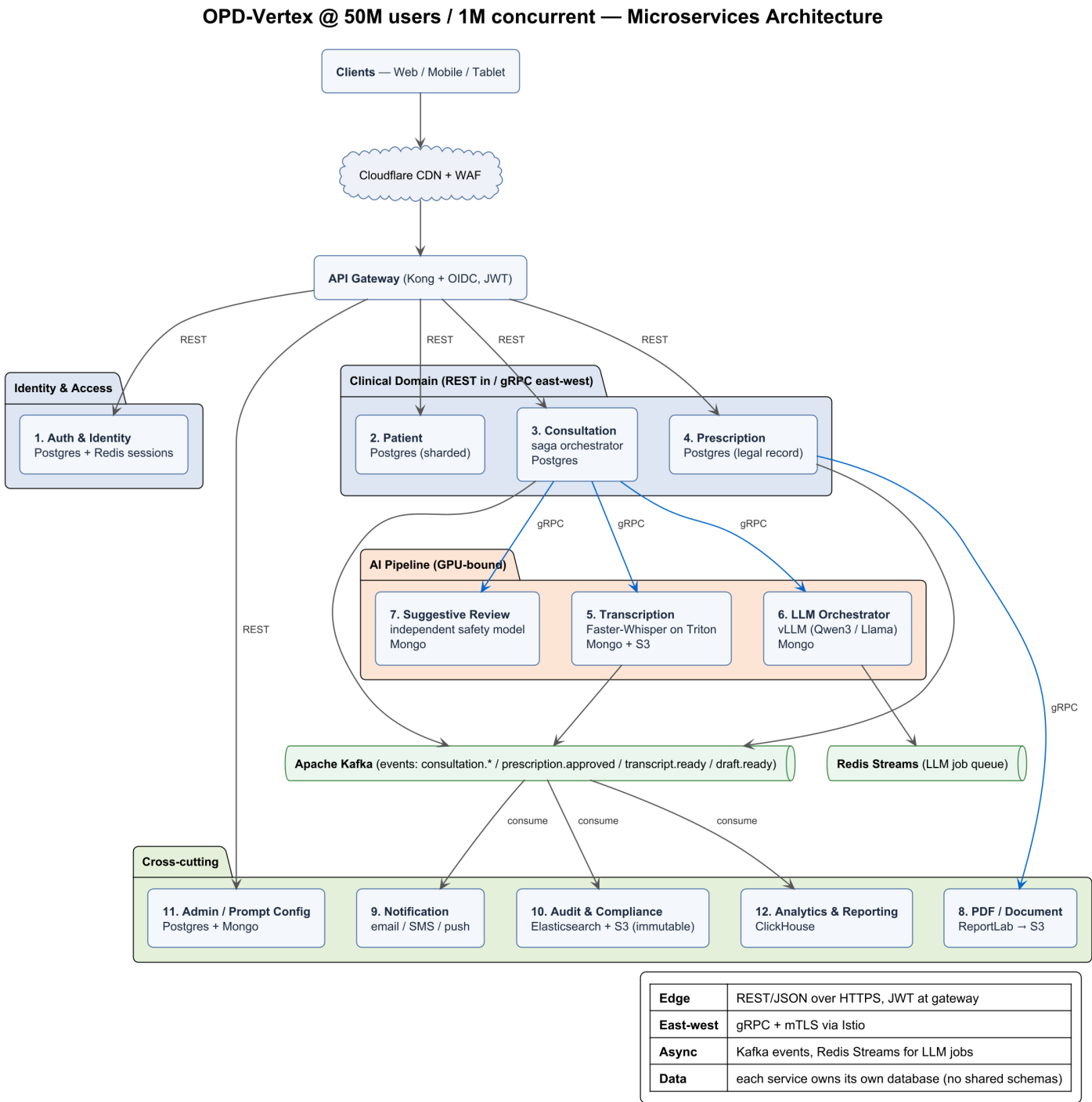
Service	Responsibility
1. Auth & Identity	OIDC, JWT, MFA, RBAC, session store.
2. Patient	Patient master data, demographics, consent. Read-heavy.
3. Consultation	Lifecycle saga orchestrator (recording -> approved).
4. Prescription	Approved, versioned prescriptions (legal record).
5. Transcription	Audio in -> Faster-Whisper on Triton GPU pool.
6. LLM Orchestrator	Prompt assembly, vLLM routing, retries, batching.
7. Suggestive Review	Independent safety / contradiction model on drafts.
8. PDF / Document	Server-side rendering + signing.
9. Notification	Email / SMS / push, consumes Kafka.
10. Audit & Compliance	Append-only log, GDPR export/erase.
11. Admin / Prompt Config	Prompts, templates, feature flags.
12. Analytics & Reporting	ClickHouse dashboards fed by Kafka.

Why split this way

- DDD bounded contexts: one service per context, one DB per service - no shared schemas.
- Independent scaling axes: GPU (Whisper, LLM) vs CPU (CRUD) vs IO (PDF, audit).
- Failure isolation: an LLM outage must not block patient lookups or PDF reads.
- Compliance: Auth + Audit are isolated and centrally enforced for HIPAA/GDPR.
- Team topology: 12 services map cleanly onto 4-6 squads.

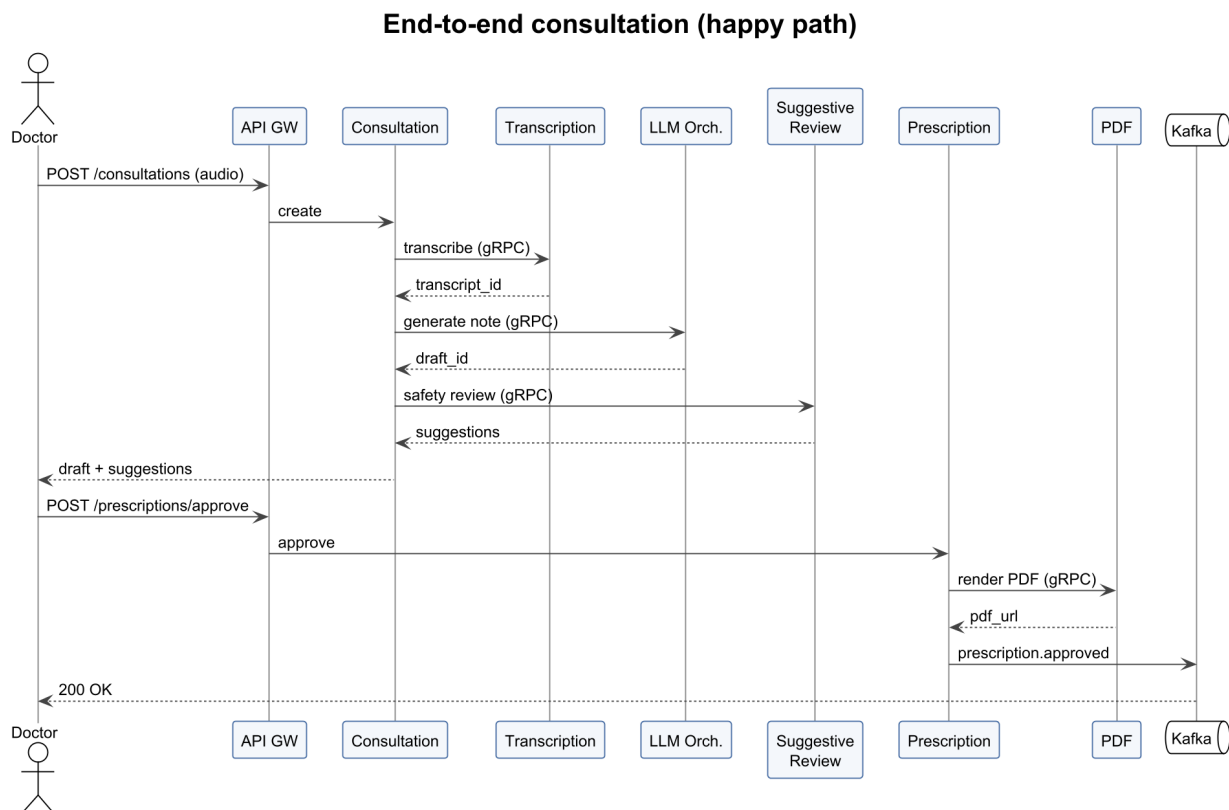
2. Architecture Diagram

External traffic is REST/JSON terminated at Cloudflare + Kong (JWT validation). Internal east-west calls are gRPC over an Istio service mesh with mTLS. Long-running and fan-out work is decoupled via Kafka events and Redis Streams (LLM jobs). Each service owns its own database; cross-service reads come from event-sourced projections, never shared tables.



Component view - colored bands group bounded contexts; data ownership annotated inside each service.

End-to-end consultation flow



3. Benefits and Trade-offs

Benefits

- Independent scaling: GPU pools scale separately from CRUD - 60-80% infra savings vs scaling the monolith.
- Resilience: Istio bulkheads + circuit breakers keep failures local; LLM outage degrades to draft-later mode.
- Faster delivery: 12 deployable units, blue/green per service, no shared release train.
- Polyglot persistence: Postgres for legal records, Mongo for drafts, S3 for blobs, ClickHouse for analytics.
- Compliance centralized in Auth + Audit, simplifying GDPR/HIPAA audits.

Trade-offs

- Distributed data consistency: sagas (Consultation orchestrates) + outbox pattern; eventual consistency on read models.
- Operational complexity: 12 services + mesh + Kafka + GPUs require strong SRE practice and observability.
- Network cost / latency: ~2-5 ms per gRPC hop; mitigated by co-locating chatty services in the same node pool.
- Local-dev friction: mitigated by Tilt/Skaffold + service stubs and contract tests (Pact).
- GPU cost: KEDA scale-to-zero + request batching in vLLM and Triton.

4. Tech Stack and Justification

Layer	Choice	Why
Runtime	Python 3.12 + FastAPI	Existing codebase; async I/O; OpenAPI native.
Internal RPC	gRPC + Protobuf	Low latency, typed contracts, streaming for audio.
Edge / Gateway	Cloudflare + Kong	Global CDN, WAF, rate limiting, JWT at edge.
Service mesh	Istio + Envoy	mTLS, retries, circuit breakers, traffic shifting.
Async bus	Apache Kafka	Durable, partitioned, replayable; fits audit fan-out.
Job queue	Redis Streams	Sub-ms enqueue for LLM jobs, consumer groups.
SQL	PostgreSQL (Patroni + Citus)	ACID for legal records; mature HA + sharding.
NoSQL	MongoDB	Flexible schema for evolving LLM drafts/transcripts.
Cache / sessions	Redis Cluster	Sub-ms read-through cache, session store.
Object store	S3 / MinIO	Cheap, durable for audio + PDFs; lifecycle to Glacier.
Search / logs	Elasticsearch + Loki	Full-text patient/audit search; log aggregation.
Analytics	ClickHouse	Columnar, sub-second on billions of rows.
LLM serving	vLLM (Qwen3 / Llama 3) on H100	Continuous batching, paged attention.
Speech-to-text	Faster-Whisper on Triton	GPU-batched ASR, streaming.
Orchestration	Kubernetes (EKS) + KEDA	Event-driven autoscaling incl. GPU pools.
CI/CD	GitHub Actions + ArgoCD + Trivy	GitOps, SBOM, image scanning.
Observability	Prometheus / Grafana / Tempo / OTel	RED/USE metrics, distributed traces, SLO alerts.
Secrets / IaC	HashiCorp Vault + Terraform + Helm	Short-lived creds; reproducible multi-region.

Deployment topology - multi-region active/active

